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Futureproof

Equity Preservation Mortgage®

Modelling v14d (Optimized)

EPM Model Review

Principal + Interest Mortgage

May 2026

Based on the Equity Preservation Mortgage (EPM) v14d (Optimized)
Model GBM (with Stochastic Drift) + Mean Reversion Equity Model
(Shevchenko 2026) 50,000-path Monte Carlo Simulation | May 2026

Contents

1. Executive Summary

The headline result of this review is a reassuring one: index-linking to the S&P500 is effective product design and the Equity Preservation Mortgage® (EPM) model produces a **strongly positive expected outcome**. Across 50,000 simulated 30-year paths, roughly **92% of mortgages fully self-fund** — the mortgage offset account grows past the mortgage loan balance, with a typical (median) surplus near $1.0M shared between Futureproof and the funder. The homeowner, for their part, draws a guaranteed income for a their selected annuity term and retains their home with its equity fully preserved in every outcome, bearing none of the investment risk.

The Payments & Risk Waterfall successfully risk manages the EPM funding mechanism, reducing:

- Index volatility
- Reference asset portfolio volatility
- Intra-term probability of deficit (PoD)
- End-of-term probability of claim (PoC)

The insurance and reinsurance layers cleanly absorb the tail risk arising in a minority of return pathways. The body of this report verifies and validates the engineering behind that result.

This report reviews the Futureproof EPM model as a piece of quantitative engineering — the stochastic processes, simulation mechanics, insurance pricing, and the Payments & Risk Waterfall. Its purpose is to document how the model works and to verify that the logic is internally consistent. A separate companion papers — "*Actuarial Analysis and Modelling Methodology*" and "*Model Assumptions and Parameter Risk*" — examine whether actuarial approach and methodology for modelling index returns are correct and that the *input parameters* are empirically defensible.

A critical distinction in this review is between Probability of Deficit (PoD) and Probability of Claim (PoC).

PoD is a balance sheet snapshot: It shows what fraction of paths are in deficit at a given point in time.

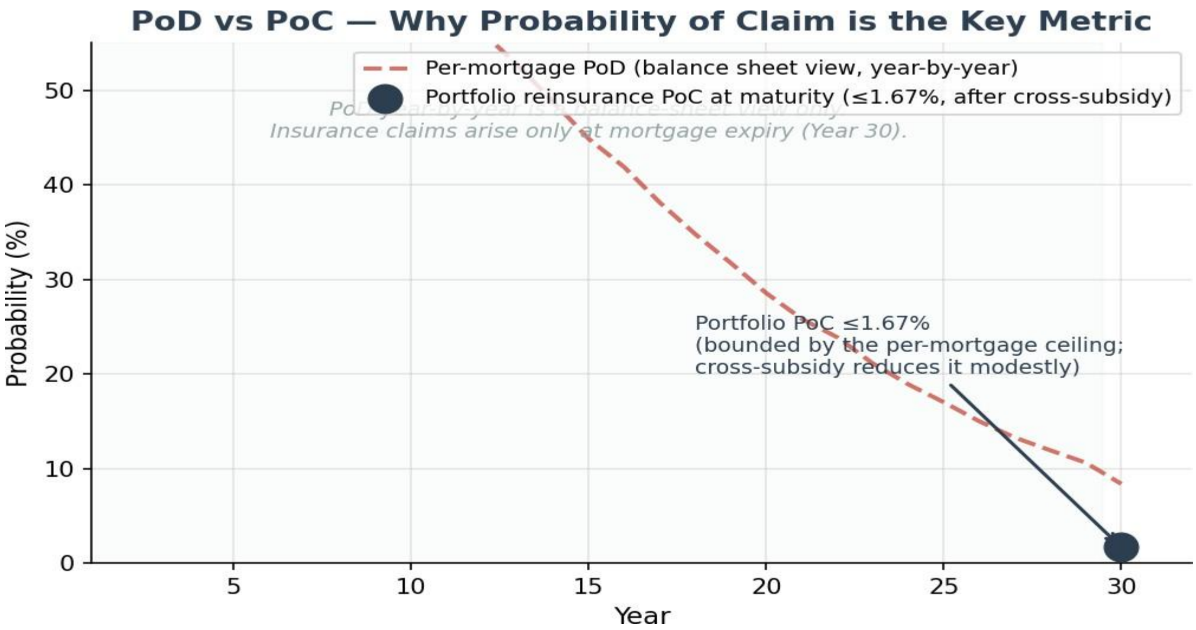
PoC is the actuarially relevant metric: It measures the probability of an actual insurance or reinsurance claim, which can only occur when a mortgage expires. PoC is the primary metric for insurance and reinsurance pricing.

All headline numbers in this report are *model-as-written* — computed at the current parameter set (μ = 9.2%, γ = 0.163, θ = 2.13%, collar cost 0.046%, asymmetric collar cap +40% / floor -20%, profit share 10% at 3-year resets). The companion Model Assumptions paper discusses how outputs shift under alternative parameter calibrations.

Key Findings

- Independent 50,000-path Monte Carlo confirms **~92% of mortgages finish in surplus** at maturity (per-mortgage PoD 8.37%, SE 0.12%), with a median surplus of ~$1.0M
- **Portfolio reinsurance PoC is bounded above by the per-mortgage LMI reinsurance PoC of 1.67% (the hard ceiling)** and sits modestly below it after the Payments & Risk Waterfall (ETF selldown + cross-subsidization from surplus mortgages). Cross-subsidization pools the ~92% of mortgages that reach maturity in surplus against the ~8% in deficit, reducing residual claims.
- Insurance claims arise only on EXPIRING mortgages. The per-mortgage LMI reinsurance PoC (1.67%) is the hard ceiling for the portfolio figure — cross-subsidization and diversification can only reduce it. (Note: the portfolio balance-sheet deficit probability is a PoD snapshot, not a claim metric.)
- Insurance claims can only arise at mortgage expiry (Year 30) — PoD before expiry is irrelevant to insurance
- Asymmetric volatility collar (cap +40% / floor -20%) manages downside exposure on 16.6% equity vol
- Stochastic OU cash rates with equity correlation of 0.30 (appropriate for 30-year horizons)
- Insurance premium based on discounted expected deficit (S3), discounted at 2.13% (cash rate mean)
- LMI covers 80% of loss; worst 20% of deficit paths (larger losses) transferred as tail risk from the per-mortgage level to the portfolio (loan book) level where portfolio reinsurance applies
- Surplus at maturity split 50/50 between Futureproof and Mortgage Funder
- Profit realisation every 3 years: 10% of running surplus drawn at each 3-year hedging reset with balance of 90% of surpluses re-invested in index reference assets held within each borrower mortgage offset account

2. PoD vs PoC — The Critical Distinction



Probabiloty of Deficit (PoD)

Probability of Deficit (PoD) is a balance sheet view — it shows whether the mortgage offset account is below the mortgage balance at a particular point in time. PoD is calculated on unexpired mortgages. PoD is naturally high in early years (58.3% at year 1) — not because anything is going wrong, but because:

- Upfront costs of insurance premiums and hedging costs reduce opening balance at start-of-loan
- The mortgage offset account is still accumulating against the freshly-drawn mortgage before compounding takes over (much as a newly-drawn mortgage is "underwater" the day it is taken out). It declines steadily to 8.37% by year 30 as the account compounds and outgrows the mortgage balance
- The typical (median) path crosses into surplus around year 14 and finishes at year 30 (end-of-term) with around $1.0M of surplus in the mortgage offset account.

Probability of Claim (PoC)

Probability of Claim (PoC) is the actuarially relevant metric. An insurance claim can **only** be made when a mortgage loan contract expires — any PoD prior to that point is irrelevant to insurance and reinsurance. This is why PoC is calculated only on expiring mortgages at end-of-term (Year 30).

Furthermore, before any insurance claim is triggered, the **Payments & Risk Waterfall** is applied:

This waterfall combines standard and accepted risk management techniques widely used in capital and insurance markets for the management of retirement assets.

The practical effect of the Payments & Risk Waterfall is to significantly reduce PoD and PoC:

- **Step 1** — Interest Holidays: Where the mortgage offset account balance for a borrower falls below its original opening balance at start-of-loan by 25% or more (the Holiday-Entry trigger), an interest holiday automatically occurs in order to: preserve offset capital; maximize the return of reference assets (the index ETFs) as the market rebounds; and maintain the positive effect of compounding returns on offset capital invested in index reference assets.
- **Step 2** — ETF Selldown: The individual mortgage's offset account holds a calculated amount of loan capital as reference assets (S&P; 500 index-linked ETFs), sized by reference to the expected return pathway. ETFs are liquidated to cover the loan cost (interest) and outstanding mortgage balance (part principal) each quarter.
- **Step 3** — Capital Buffer: A further portion of loan capital is held in the individual mortgage's offset account, also as reference assets (S&P; 500 index-linked ETFs). The capital buffer is sized by reference to historical index drawdown in any 12-month period (15–20%). These ETFs are liquidated if needed, and only after depletion of the Step 2 selldown.
- **Step 4** — Continuous Dynamic Hedging: A capital-market risk-management solution. An index collar overlay using a laddered 140/80 hedge — put options against long-term call options on the portfolio of ETFs held in the individual mortgage offset account. The 80/140 hedge is reset each 1, 3 or 5 years, always limiting the loss to 20%.

[Steps 1 to 4 are continuously cycled until maximum interest-holidays limits are reached and then all reference assets (Index-lined ETFs) are exhausted as a result of progressively being sold down to pay loan costs.]

- **Step 5** — LMI: An insurance-markets risk solution. Every mortgage is covered by an LMI policy insuring the lender. The policy is subject to a top cover limit (in $) representing the worst 20% quantile of return pathways (the tail risk) calculated for each mortgage. Only the first 80% of loss triggers an insurance claim. This is the true Insurance PoC (Probability of Claim); PoC at end-of-term (Year 30) is kept within a target range of □10%. For insurance of this primary LMI risk (at a per mortgage level) PoC is kept in the target range of ≤ 2%.
- **Step 6** — Portfolio Cross-Subsidization: A risk-pooling solution. The LMI tail risk at end-of-term is transferred from the individual mortgage to the portfolio (loan book) level. This pooling mechanism uses surpluses from performing unexpiring mortgages in a funder's loan book to offset deficits of underperforming expired mortgages. Cross-subsidization between mortgages within a loan book aims to reduce overall PoC on Portfolio Reinsurance by a target range of 2-3%.

[Portfolio cross-subsidization is limited to within each funder’s loan book. One funder does not cross-subsidize the mortgages of another funder. For each funder, cross-subsidization between their expiring and expired mortgage is a zero-sum game. The deficit of a an expired mortgage subsidized intra-term by unexpired the surpluses of unexpired mortgages in their loan book would have, otherwise, simply been carried forward and reduced the end-of-term mortgage surplus by an identical amount.]

- **Step 7** — Portfolio Net Deficit: An insurance-markets risk solution. Only the residual deficit (calculated only on expiring mortgages, not all mortgages) after Steps 1 through 6 triggers a reinsurance claim. This is the true Reinsurance PoC: Reinsurance PoC at end-of-term is kept within the target range of ≤ 4-6%.

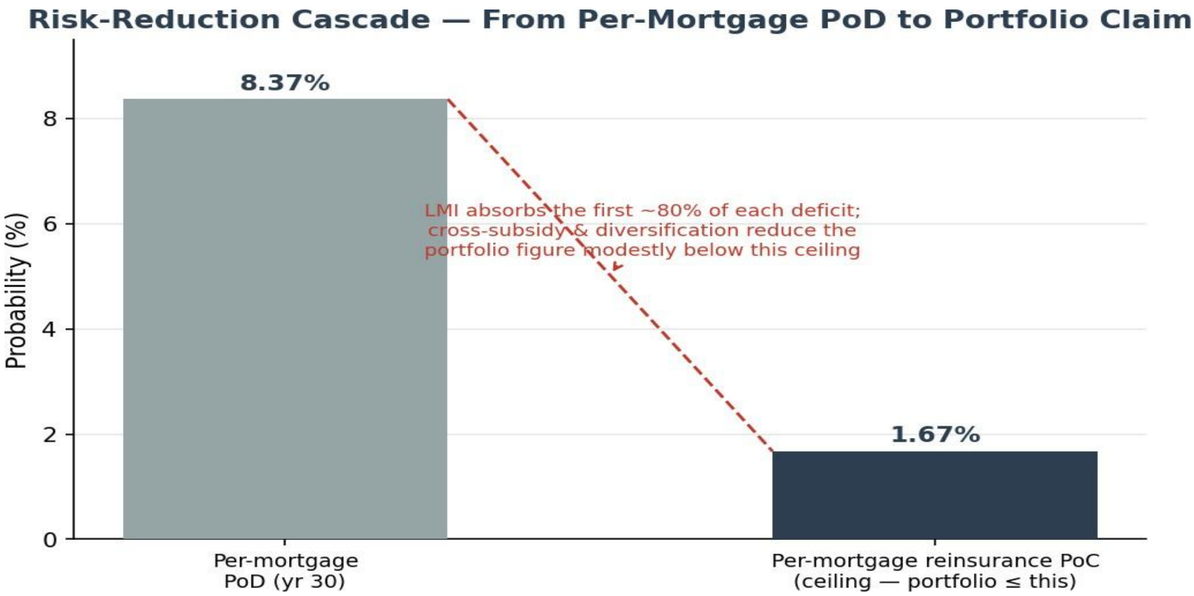
Results:

> The Payments Waterfall (ETF selldown plus cross-subsidization from surplus mortgages) reduces the probability of a reinsurance claim from the per-mortgage ceiling (1.67%) to a modestly lower portfolio level:

LMI tail risk (20% quantile) is transferred from the per mortgage level to the portfolio (loan book) level where loss is covered by portfolio reinsurance, At the portfolio level, the Payments & Risk Waterfall then pools the surpluses of the ~92% of mortgages that reach maturity in surplus against the deficits of the ~8% that underperform; only the residual after this cross-subsidy can trigger a portfolio reinsurance claim. The cross-subsidy is the primary risk-mitigation mechanism, and it is powerful. (The portfolio balance-sheet deficit probability is a PoD snapshot of the whole book, not a claim metric — see the Portfolio Probability of Claim section.)

From a reinsurance perspective, the portfolio reinsurance PoC (below 1.67%) sits below the per-mortgage reinsurance PoC (1.67%, the hard ceiling) and well below the per-mortgage PoD (8.37%).

3. Portfolio (Loan Book) Probability of Claim



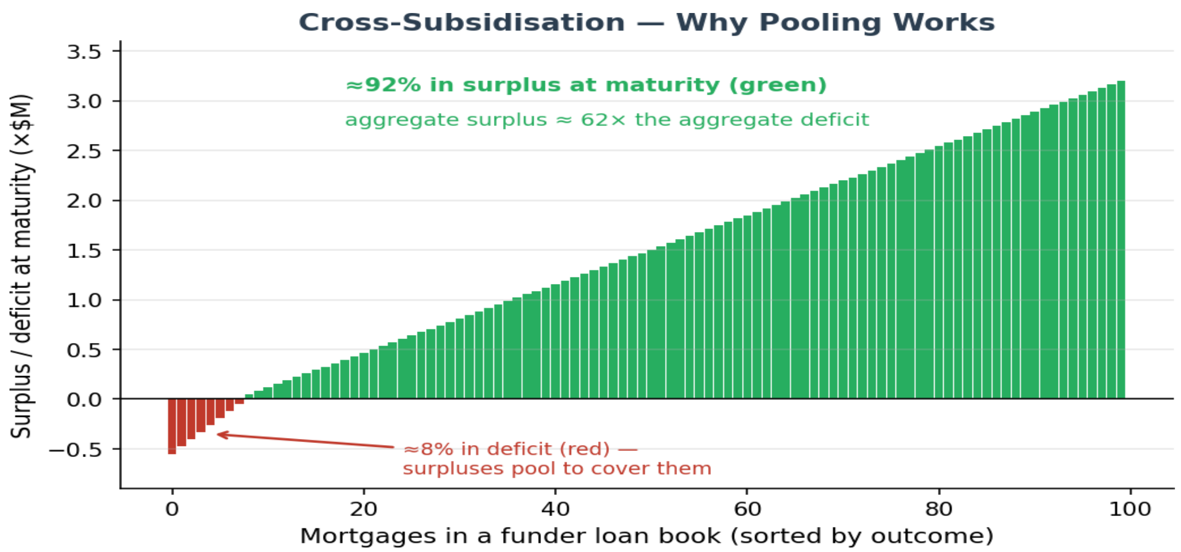
The cascade above shows how the Payments & Risk Waterfall reduces risk: the LMI layer absorbs the first ~80% of each deficit (taking per-mortgage PoD down to the per-mortgage reinsurance PoC ceiling), and portfolio cross-subsidisation and diversification then pool surpluses against deficits — taking the portfolio reinsurance PoC modestly below that ceiling.

Metric	Value	Notes
Per-mortgage PoD (yr 30)	8.37%	One mortgage in deficit at maturity (balance-sheet view)
Per-mortgage LMI insurance PoC	8.37%	Covers 80% losses up to top-cover limit (20% quantile)
Per-mortgage LMI reinsurance PoC	1.67%	Deficit beyond the LMI top-cover boundary — the hard ceiling for the portfolio figure
Portfolio reinsurance PoC (after cross-subsidy)	< 1.67%	Bounded by the per-mortgage ceiling; diversification and cross-subsidization reduces it significantly

What this means. The per-mortgage reinsurance PoC (1.67%) is the hard ceiling for the portfolio figure, which sits far below the per-mortgage PoD (8.37%). Two related figures describe why the pool is always deep, and they should not be confused:

- **At maturity (the only point a claim can arise): ~92% of mortgages are in surplus and only ~8% in deficit** — this is the 8.37% per-mortgage PoD. By year 30, P&I; amortisation has shrunk the loan and the offset account has had three decades to compound, so the large majority finish well in surplus.
- **Across the full 30-year horizon, ~85% of return pathways are positive and ~15% negative** — a broader figure that also counts paths dipping into deficit at some point before recovering. It explains why, at any point in time, a funder's loan book holds far more surplus mortgages than deficit ones (otherwise the S&P; 500 would not appreciate over the long run).

Either way, the surpluses pool to cover the deficits, so only the residual after the Payments & Risk Waterfall can trigger a claim. The per-mortgage LMI reinsurance PoC is the hard ceiling — cross-subsidisation can only reduce claims, never increase them. It is the dominant risk-mitigation mechanism, and it is powerful.


Cross-Subsidisation — Why Pooling Works
≈92% in surplus at maturity (green)
aggregate surplus ≈ 62× the aggregate deficit
≈8% in deficit (red) —
surpluses pool to cover them
Surplus / deficit at maturity (×$M)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
20
40
60
80
100
Mortgages in a funder loan book (sorted by outcome)

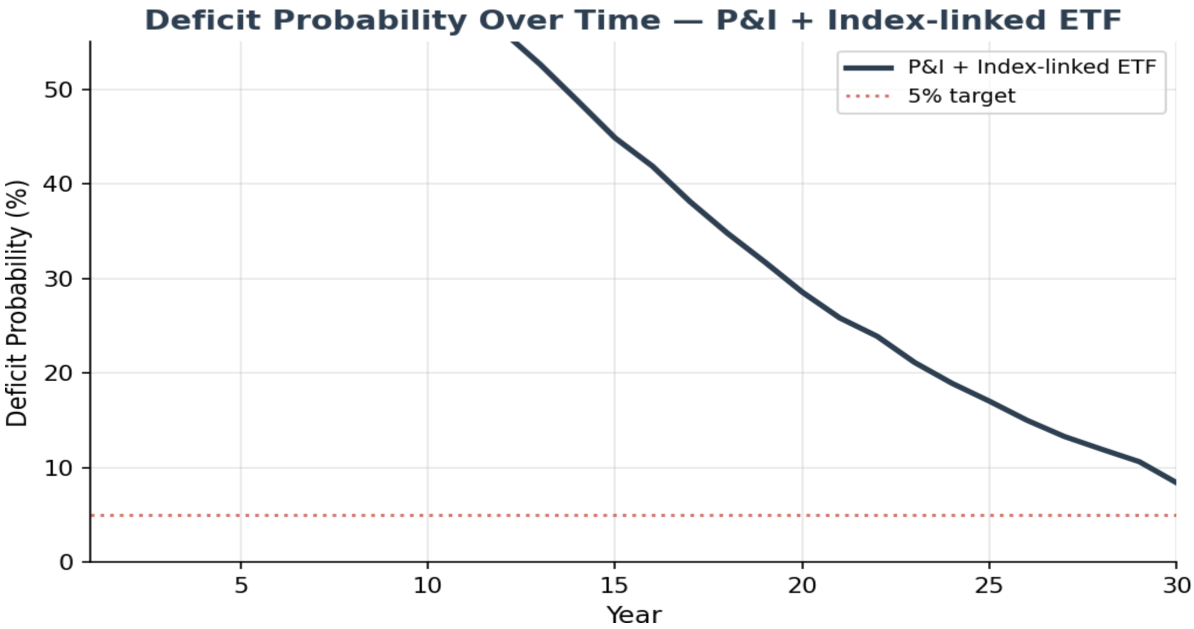
4. Portfolio PoC — Provenance and the PoD/PoC Distinction

Three distinct "deficit probability" metrics circulate in EPM analysis and are easy to conflate. The table distinguishes them. The critical point: a **portfolio deficit-probability snapshot** (PoD) is not a claim probability (PoC). Insurance claims arise only on expiring mortgages, after the Payments Waterfall.

Metric	What it measures	Type	Value
Per-mortgage PoD at maturity	Single mortgage: mortgage offset account < loan balance at year 30	PoD	8.37%
Per-mortgage LMI reinsurance PoC	Single mortgage: deficit exceeds LMI top-cover boundary at maturity	PoC (ceiling)	1.67%
Portfolio reinsurance PoC	Residual reinsurance claim on expiring mortgages after cross-subsidy	PoC (the figure to quote)	67%
Portfolio balance-sheet deficit prob.	Aggregate portfolio balance < 0 at a point in time (xlsm Portfolio sheet)	PoD (NOT a claim metric)	≈ per-mortgage PoD

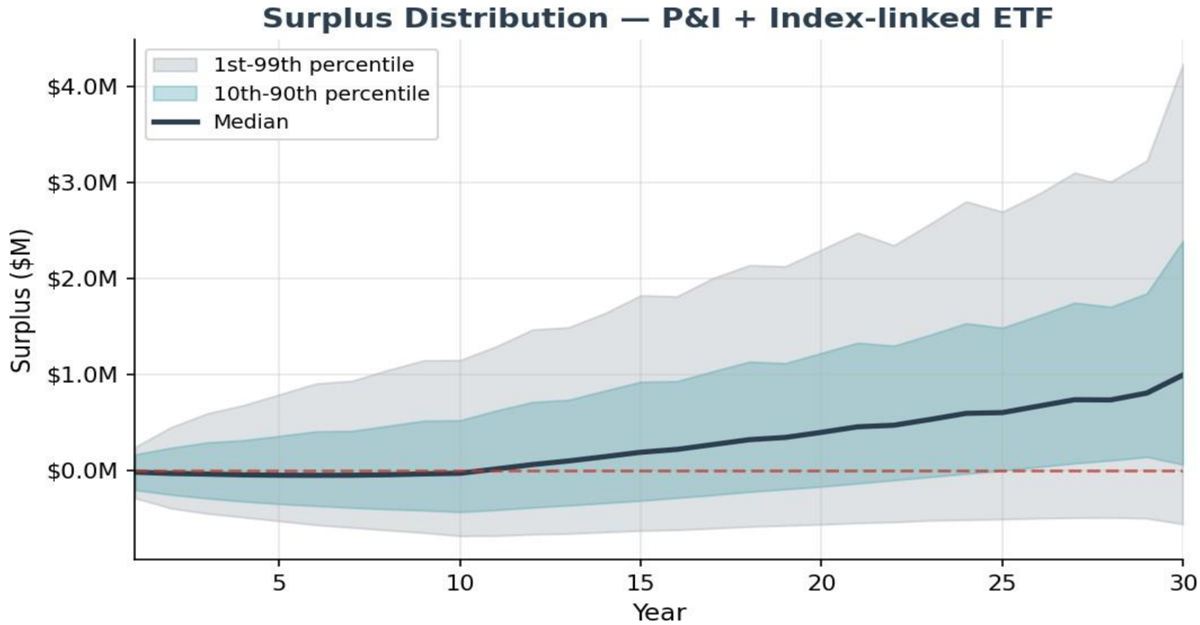
What the portfolio PoC is NOT. The xlsm Portfolio sheet, labelled "prob of portfolio deficit", is the probability the aggregate book balance is negative at a point in time — a PoD snapshot of the whole portfolio (expiring plus unexpiring mortgages). It is not a claim metric: claims arise only on expiring mortgages, after cross-subsidy. That figure tracks the per-mortgage PoD and must not be quoted as portfolio PoC.

5. Deficit Probability (PoD) Over Time — Balance Sheet View



PoD shows the fraction of paths in deficit at each point in time — a useful balance sheet view but **not** the probability of an insurance claim. PoD starts at 58.3% in year 1 and declines to 8.37% by year 30. The rapid decline after year 10 reflects P&I; amortisation — as the loan shrinks, the investment needs only modest returns to exceed the declining balance. Note: PoD at any point before mortgage expiry is irrelevant to insurance and reinsurance — only PoC on expiring mortgages matters.

6. Surplus Distribution — Fan Chart



The fan chart shows the distribution of surplus (investment balance minus mortgage balance) over the 30-year tenure. The median surplus reaches $993,211 at maturity. The 10th-90th percentile band widens over time but remains predominantly positive after year 15.

Important: Surplus at maturity is split 50/50 between Futureproof and the Mortgage Funder — it does not go to the borrower. Additionally, partial profit realisations occur every 3 years on hedging program re-sets (10% of surplus drawn). These periodic realisations are factored into the model.

7. Holiday Mechanism Analysis

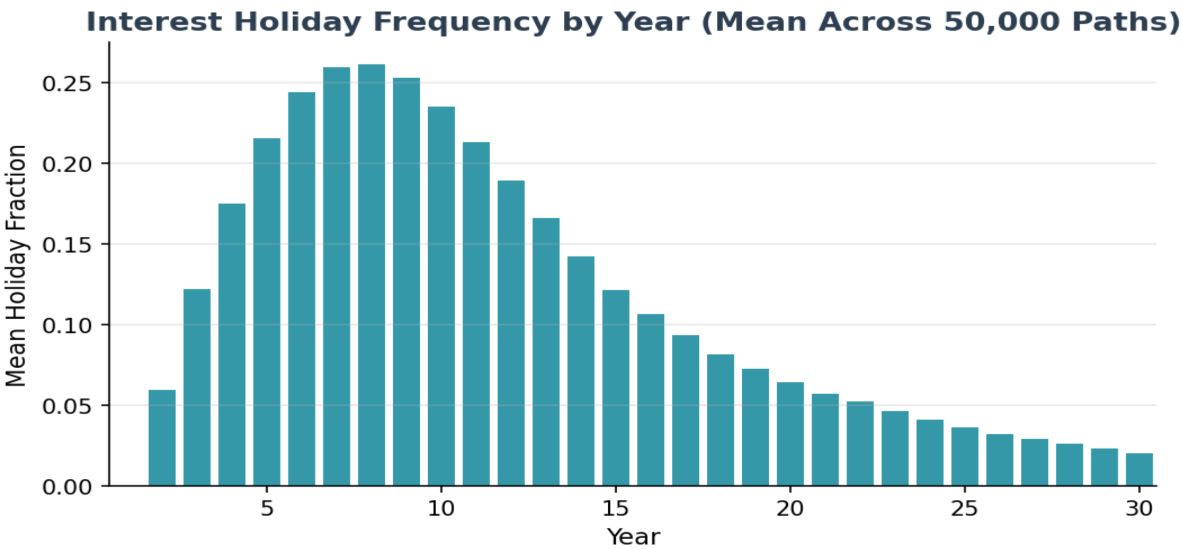
The interest holiday mechanism is a protective feature that temporarily pauses payment of loan costs (interest charges and part-principal repayments) when the offset account falls below a threshold (0.75× the initial offset account balance at start-of-loan).

Paused interest is deferred, not accrued and compounded — it remains charged as simple interest.

An interest holiday ceases when the mortgage offset account balance is restored over time to the holiday-exit threshold (currently 145.8% of the original offset balance at start-of-loan). This iterative process of ETF sell-down versus interest holidays is continuous through the loan lifecycle — managing index volatility and economic cycles — until all ETFs are exhausted, which would result in a deficit for that mortgage.

Per-Year View — In Most Years the Median Mortgage Needs No Holiday

In any given year of the 30-year term, **the median mortgage requires zero interest holidays** (the per-year holiday rate stays below 50% throughout). Individual paths that experience adverse early-year returns can enter holiday mode. Holiday frequency peaks in years 2-8 during the annuity period, dips after annuity cessation at year 10, then gradually rises again in later years as fee drags accumulate on underperforming paths.



The chart above shows the mean holiday fraction by year across all 50,000 paths. Holidays concentrate in the early years — while the mortgage offset account is still building against the freshly-drawn mortgage — peaking at roughly a quarter of paths around year 7-8, then fading to about 2% by maturity as the account compounds.

The median path is never on holiday in any year, and over the full term more than half of mortgages never enter a holiday at all (see below). A holiday never affects the borrower — they keep receiving their annuity throughout; it is an internal safety valve, not a sign of borrower distress.

Year Range	Median (per year)	P90 (per year)	P99 (per year)	Interpretation
Years 1-10	0	1	1	Early-years safety valve while the account builds; median path takes none
Year Range	Median (per year)	P90 (per year)	P99 (per year)	Interpretation
Years 11-20	0	0-1	1	Fading as the account compounds
Years 21-30	0	0-1	1	Rare — only on weaker paths

Cumulative View — Total Holidays Over 30 Years

Viewed cumulatively over the full 30-year term, **more than half of mortgages (55%) never enter a holiday at all**, and the typical (median) mortgage takes none. Across the whole book the average is just 3.5 holiday-years over 30 years. Holidays are not defaults — the borrower keeps receiving their annuity, deferred interest is repaid as the mortgage offset account recovers, and usage concentrates on the minority of paths with adverse early-year returns:

Metric	Value (50,000-path)	Interpretation
Paths with zero holidays	55%	Never require a holiday over 30 years
Paths entering holiday	45%	Use the holiday mechanism at least once
Mean holiday-years (all paths)	3.5	Average across the full distribution
Mean holiday-years (conditional)	~7.7	Average for paths that do enter holiday

Per-year holiday rates use the 50,000-path simulation; the full cumulative percentile distribution (P75/P90/P99 of total holiday-years) awaits a dedicated holiday-analysis run on the current model.

Holiday Repayments & Insurance Coverage

Critically, the number of interest holidays must be understood in the context of **subsequent holiday repayments**. When market conditions improve following downturns — as they historically do over the long term — the deferred interest is repaid from the recovering mortgage offset account. The holiday mechanism is designed as a temporary pause, not a permanent deferral.

If the net result of holidays and subsequent repayments is that any amount of loan cost remains outstanding at end-of-term, it is covered by **insurance up to the policy top cover limit** (LMI, covering 80% of deficit paths). Any balance beyond the LMI top cover limit is covered by **portfolio reinsurance**, where the Payments & Risk Waterfall (ETF selldown, cross-subsidization from surplus loans) is applied before any reinsurance claim is made.

Holidays are a temporary pause, not a permanent deferral: when markets recover, the deferred interest is repaid from the rebuilding mortgage offset account. Any deferred interest still outstanding at maturity is rolled into the year-30 deficit calculation and is therefore already captured in the per-mortgage PoD and the insurance/reinsurance figures elsewhere in this report — **it is not an additional uncovered exposure**.

8. What the Model Gets Right

- **Investment structure:** Loan proceeds held back by the lender in the borrower's mortgage offset account are held as index reference asset managed (pursuant to a mandate from the lender) by BlackRock using passive index-linked ETFs.
- **Two-layer volatility reduction**: BlackRock Volatility Control (vol control layer) then SpiderRock continuous dynamic hedging via asymmetric collar (cap +40% / floor -20%). Modelled as cap/floor (1.4/0.8) on quarterly returns.
- **Stochastic interest rates:** The OU mean-reversion model (speed=0.24, vol=1.22%) is appropriate for Australian cash rates.
- **Equity-rate correlation at 0.30:** Over 30-year horizons, the correlation between cash rates and equity/investment returns is marginally positive (typically 0.1-0.3). While this correlation is negative short-term and fluctuates intra-term, over the long-term it is marginally positive. The 0.30 assumption is appropriate for the 30-year product horizon.
- **Deterministic house prices (by design):** The EPM is not a shared appreciation mortgage. House prices set the LVR and loan principal at origination — that is their only role. There is no need to model house price movements as they do not affect the reference asset portfolio nor loan mechanics.
- **Run-off mechanism (no early exits):** The EPM mortgage offset account has a unique run-off mechanism — no voluntary prepayments or early terminations are permitted until all loan cost has been paid from the offset account or, alternatively made certain at end-of-term by mortgage insurance (whichever occurs first). This eliminates early exit risk that affects traditional mortgages.
- **Insurance pricing:** LMI and tail risk (reinsurance) premiums are paid upfront at mortgage origination. Premium is calculated on the discounted expected deficit (S3), with LMI covering 80% of deficit paths and the worst 20% of deficit paths (larger losses) transferred as tail risk to reinsurance.
- **Payments & Risk Waterfall:** The portfolio-level waterfall (ETF selldown → cross-subsidization → net deficit) dramatically reduces the actual Probability of Claim from individual PoD levels.
- **Profit realization:** 10% of running surplus drawn every 3 years — correctly modelled.

9. Insurance Premium Analysis

Premium Calculation Methodology

The insurance premium is calculated on the **discounted expected deficit (S3)**, not on raw deficit values. This discounting is necessary because no reinsurance claim can be made until mortgage expiry (Year 30). Both LMI and reinsurance premiums are paid upfront at mortgage origination. The discount rate of 2.13% corresponds to the long-run cash rate mean (MLE estimate).

Two-Layer Insurance Structure

- **LMI (Lenders Mortgage Insurance):** Covers 80% of deficit paths (smaller losses). Premium is based on the discounted expected deficit at mortgage expiry. Both the LMI and tail risk premiums are paid upfront at mortgage origination.
- **Tail Risk (Reinsurance):** Covers the worst 20% quantile of losses. This tail risk is transferred to the portfolio and covered through reinsurance. Critically, the Payments & Risk Waterfall is applied before any reinsurance claim is made — and claims can only arise at mortgage expiry (Year 30). This is why the portfolio reinsurance PoC sits modestly below the per-mortgage reinsurance PoC of 1.67%, which is the hard ceiling — cross-subsidization and diversification can only reduce it.

Metric	Value	Notes
Discounted Expected Deficit (S3)	$143,274	Basis for premium calculation
Discount Rate	2.13%	cash rate mean yield
Fair LMI Premium (PV)	$10,168	SE: $120; covers 80% of deficit paths
Fair + Loading (50%)	$15,253	1.27% of max loan
Tail Risk Premium	$1,827	Worst 20% of deficit paths (larger losses); covered via reinsurance
Individual PoD at Yr 30	8.37%	Balance sheet view (before waterfall)
Per-mortgage reinsurance PoC	1.67%	Deficit beyond LMI boundary — hard ceiling for portfolio PoC
Portfolio reinsurance PoC	≤ 1.67%	Bounded by the per-mortgage ceiling; modestly lower after cross-subsidy

10. Actuarial Assessment

Is the Logic Sound?

Yes. The model is a well-constructed Monte Carlo simulation with appropriate stochastic processes for equity returns (GBM with mean reversion per Shevchenko 2026) and interest rates (Ornstein-Uhlenbeck with exact discretisation). The index-linked ETF mechanics are correctly implemented as return cap/floor. The P&I; amortization schedule is standard. The Payments & Risk Waterfall correctly models the portfolio-level risk mitigation that reduces PoC relative to PoD. The simulation converges cleanly at 50,000 paths (SE 0.12% on PoD at base case).

The logic of the model is therefore sound as a piece of quantitative engineering. This is a separate question from whether the *inputs* (μ, γ, θ, collar cost, margin) are the right inputs — which is addressed in the companion Model Assumptions paper.

Simulation precision resolved:

- Independent 50,000-path Monte Carlo confirms 8.4% PoD (SE: 0.12%) and fair premium of $10,168 at base-case parameters — actuarial-grade precision. The production spreadsheet tool now runs both per-mortgage and portfolio simulations at 50,000 paths for internal decisioning.

Are the Numbers Correct?

The outputs are internally consistent with the model's inputs and methodology. Whether the inputs themselves are correctly calibrated is examined in the companion paper; the numbers below describe the model-as-written at the base-case parameter set.

- **PoD trajectory:** 58.3% at year 1 declining to 8.37% at year 30 — consistent with the mortgage offset account compounding past the mortgage balance over the term (60.0% effective LVR with 3.45% cost drag).
- **Portfolio reinsurance PoC:** sits modestly below the per-mortgage reinsurance PoC of 1.67% (the hard ceiling) after cross-subsidization and diversification. Cross-subsidization pools surplus loans against deficit loans and is the dominant risk-mitigation mechanism.
- **Insurance premium:** Based on discounted expected deficit (S3), discounted at 2.13% over up to 30 years. Premiums paid upfront at mortgage origination; claims deferred until loan expiry.
- **Surplus split:** 50/50 between Futureproof and Mortgage Funder, with 10% partial realization every 3 years. This is correctly factored into the surplus trajectories.
- **Parameter-sensitivity (companion paper):** The Model Assumptions paper documents how outputs shift under alternative parameter calibrations. The model is correct; the inputs drive the range.

11. Summary & Recommendations

Model Strengths

- Simulation precision achieved — 50,000 paths, SE 0.12% on PoD, SE $120 on premium
- Correct distinction between PoD (balance sheet) and PoC (insurance claim on expiring mortgages)
- Payments & Risk Waterfall keeps the portfolio reinsurance PoC modestly below the per-mortgage reinsurance PoC of 1.67% — the hard ceiling
- Run-off mechanism eliminates early exit risk (unique to EPM)
- Deterministic house prices correct for this product (not a shared appreciation mortgage)
- Stochastic interest rates with exact OU discretisation; Cholesky-correlated equity shocks
- Insurance premiums based on discounted expected deficit, paid upfront at mortgage origination

Model-Level Enhancement Priorities

- [DONE] Simulation paths increased to 50,000 — actuarial-grade precision achieved
- [DONE] PoC vs PoD distinction clearly established
- [DONE] Production spreadsheet now runs both per-mortgage and portfolio simulations at 50,000 paths
- [HIGH] Re-anchor collar cost input on Black-Scholes (0.30% minimum) — current 0.046% is inconsistent with current AU rates
- [HIGH] Re-anchor θ (cash rate mean) on AU MLE, not US Fed Funds MLE — move to 3.0% minimum
- [MEDIUM] Add scenario switcher (base / realistic-central / adverse-plausible) to the model UI
- [MEDIUM] Add wholesale margin sensitivity analysis directly in the model
- [LOW] Add multi-region parameter sets (AU, NZ, UK, US)

Product Recommendations

- Maintain P&I; as the default mortgage product offering — the single most impactful risk reduction lever
- The $30K annuity on $1.5M eligible house value is well-calibrated for current base-case parameters (2% of home value p.a.)
- Report PoD as a range (base / realistic / adverse) in all internal and external materials — not a point
- Size capital and reinsurance attachment to the adverse-plausible leg, not the base case

12. EPM vs Current Investment Landscape

To contextualise the EPM risk profile, we compare its base-case metrics against established benchmarks across traditional mortgages and rated credit instruments. Stress-scenario comparisons (realistic-central / adverse-plausible) are provided in the companion Parameter Risk paper.

Metric	EPM (base case)	Prime AU Mortgage	Moody's AA Bond
Cumulative PoD (30yr)	8.37%	3–5%	~1.5%
Loss severity (LGD)	~10·12%	20–40%	40–60%
Expected loss (30yr)	~1.00%	0.6–1.25%	0.6–0.9%
Claim timing	Year 30 only	Any year	Any year
Prepayment / lapse risk	None (run-off)	Significant	Moderate

- **Structural risk driver is fundamentally different.** Traditional credit risk stems from borrower inability to pay. EPM risk stems from 30-year index underperformance relative to interest rates. These drivers are largely uncorrelated, so the EPM is a natural diversifier in a traditional credit portfolio.
- **Claim concentration at maturity.** Traditional mortgage credit incurs claims throughout the loan life; the EPM produces claims only at year 30. This concentrates capital-and-reserve requirements at maturity but eliminates intra-life prepayment and lapse risk entirely.
- **Loss severity is structurally bounded.** The ~10·12% conditional severity reflects the structural property of the EPM — at deficit, the loss is the gap between the mortgage offset account and the mortgage balance, not the full mortgage balance as in a traditional foreclosure. This severity is materially lower than corporate-bond or RMBS subordination losses.
- **Stress sensitivity to be addressed separately.** All base-case figures above use the model's central parameter calibration. The accompanying Parameter Risk paper quantifies outcomes under realistic-central and adverse-plausible parameter stresses.

13. How the EPM Works — Equity Return with Mortgage-Like Risk

There are two design principals at the core of the EPM design thesis:

- A borrower must not carry risk in retirement (market or investment risk, capital risk, interest rate risk or product risk)
- Index-linking the EPM converts long-term equity returns into a mortgage-like risk profile, delivering equity-like upside with bond-like downside protection

Index-linking only works with long-duration mortgages (20–30 years to smooth index volatility and achieve long-term total return performance): economic cycles are never 30 years (interest-rate cycles ≈ 2–3 years, mild/moderate downturns ≈ 3–5 years, black-swan events such as the GFC ≈ 5–7 years). Long duration is key to the successful design and effectiveness of the EPM.

Why index-linking gives an equity-like return

The EPM's offset account is invested in broad candidate indices (strongly correlated to the S&P; 500). The S&P; 500 remains the dominate index because:

- The correlation of all candidate indices to the S&P500 moves towards 1 in an economic downturn
- 73% of mortgage offset account capital is held in equities index assets
- 74% of those equities index assets are U.S. companies
- Modern S&P500 index data goes back more than 50 years
- Depth of capital
- Market liquidity
- Reliable historical performance data (returns and volatility)
- Deep & liquid traded options market to support hedging.

Importantly, the index itself is a risk-levelling or risk-pooling mechanism — only the top 500 capitalized US companies are included, and the S&P; committee removes/adds companies that under/over-perform. This protects against poor performers weighing down the index, ensures emerging technologies are captured (such as AI), and weeds out sunset industries over time.

The lender's capital behaves like a fully funded index exposure:

- Long-run expected return of the index (≈10.8% real; ≈10% forward-looking).
- Diversified, equally-weighted, low-cost exposure.
- Passively managed (not chasing alpha) — letting the long time horizon do the heavy lifting.
- No single-stock or manager risk.
- No leverage, no margin calls, no behavioral timing errors.

The lender is "lending" in the traditional mortgage sense — but the offset account capital is funding an index position whose terminal value repays the loan.

This is why the return profile looks like an equity sleeve inside a mortgage wrapper.

Why the risk is mortgage-like (not equity-like)

The EPM's structural protections collapse the lender's downside to something closer to short-duration mortgage risk, not equity drawdown risk:

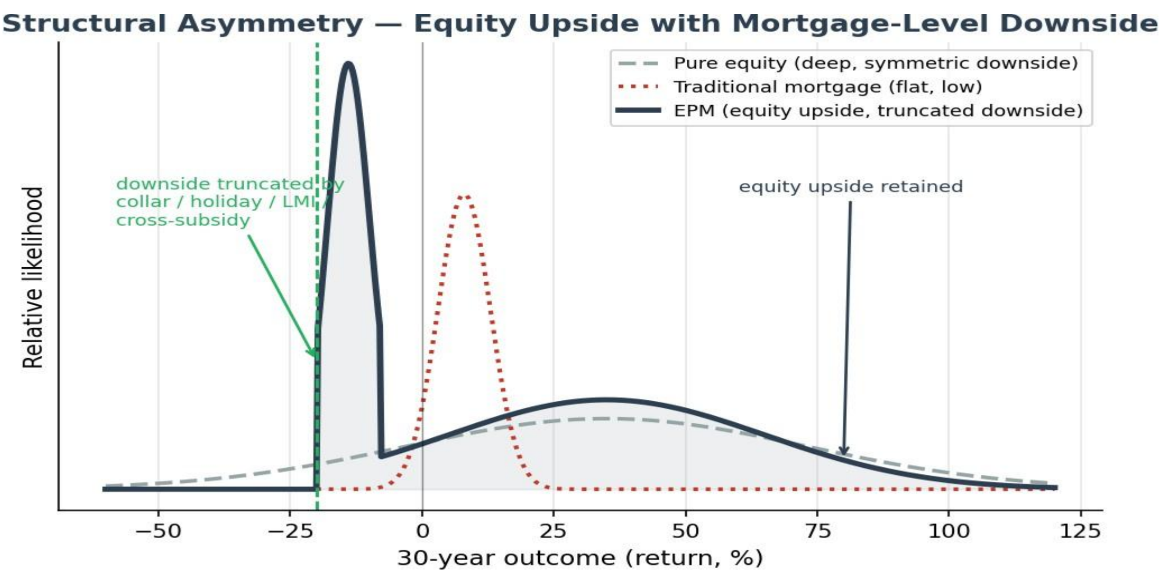
1. No borrower credit risk ? no serviceability requirement, no default, no bad debt, no forced sale, no impairment provisioning. Risk is equity-market underperformance, not borrower repayment behaviour.

- **2. No mark-to-market loss** — the offset account remains invested for the loan lifecycle, the loan is not callable, the lender is repaid at early termination or at maturity, and early termination triggers run-off mode rather than liquidation. This eliminates the classic equity risk of being forced to sell at the bottom.
- **3. Short effective duration** — annuity payments are funded separately via progressively drawn loan capital; interest and part-principal are funded by progressive sell-down of ETF reference assets; and as ETFs are sold down, the unsold ETFs continue accruing and compounding. The lender's capital behaves like a short-duration, self-amortizing asset, not a 30-year mortgage.
- **4. Capital certainty at termination** — even on early termination the lender receives full loan principal back: no loss crystallization, no negative equity, no exposure to the property market.
- Security shifts from property to the offset account capital, and the reference assets held are highly liquid.

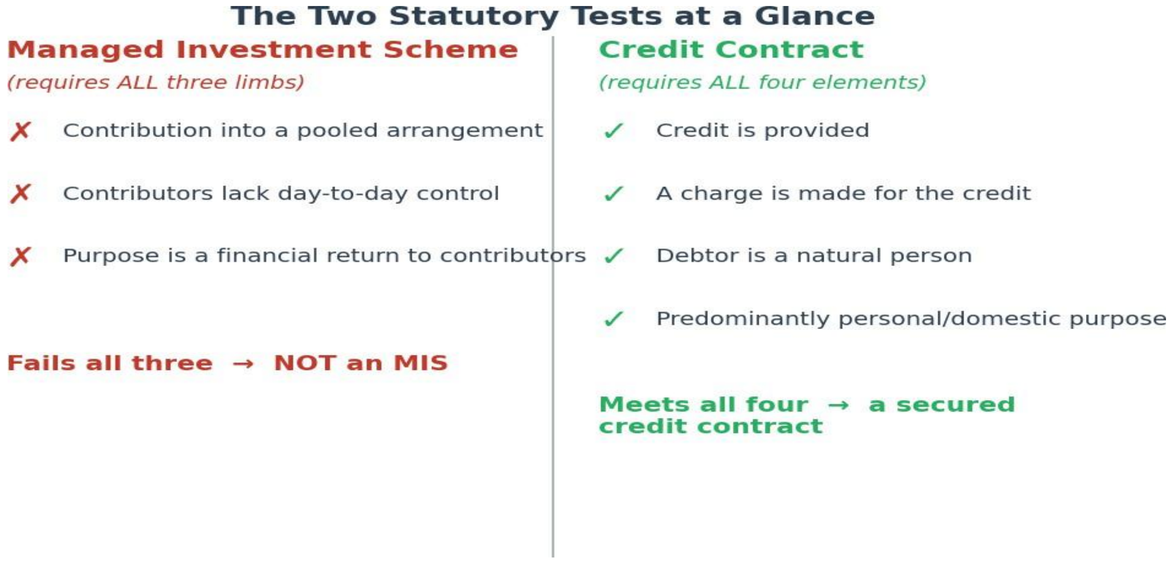
Putting it together — the payoff profile

Component	Behaves like	Why
Return	Equity index	Offset account tracks index growth
Risk	Short-duration mortgage	No credit risk, no MTM loss, no forced liquidation, capital certainty

By linking the mortgage offset account to a diversified equity index, the EPM converts long-term equity growth into a predictable, mortgage-like risk profile. The lender captures equity-style returns while avoiding borrower credit risk, market-timing risk, and mark-to-market losses — producing a structurally asymmetric payoff: equity upside with mortgage-level downside.



14. Legal Character — A Mortgage, Not a Managed Investment Scheme



Product Design Strengths

- Long loan durations (20–30 years).
- Secured by a registered first mortgage over unencumbered residential property.
- Variable or adjustable-rate mortgage.
- Recommended retail pricing 3.5–4% above cash rate (priced halfway between a forward and a reverse mortgage).
- Each EPM is a single mortgage loan account with two linked sub-accounts (Annuity Capital Account and Mortgage Offset Account).
- No compounding interest; no negative-equity risk.
- Loan cost paid via the offset account from index appreciation over the long horizon, with any deficit on a minority of mortgages cross-subsidised by surpluses of the majority of unexpired mortgages, and the EPM fully insured for any residual loss.
- Under the Mortgage Agreement the borrower is not liable to pay loan costs (interest and principal repayment); all risk is shifted to the lender and funder (then de-risked and insured). It is not a financial-advice-led product.
- Borrower receives up to 2% of home value p.a. as tax-free annuity income for 5–15 years. Credit risk has been replaced by long-term (30-year) index risk.

The structure, rights, and economic substance of the EPM are those of a single mortgage-based financial instrument.

The EPM is a secured credit contract, not a Managed Investment Scheme (MIS), because it satisfies the legal definition of credit and fails the statutory tests for an MIS.

Why the EPM fails the MIS definition

An MIS requires all three limbs: (i) people contribute money or money's worth to acquire rights in a pooled arrangement; (ii) contributors do not have day-to-day control; (iii) the purpose is to produce a financial benefit for contributors. The EPM fails all three: there is **no contribution into a common enterprise** (the borrower receives credit secured by a mortgage — there is no pooling of borrower assets, no collective investment, no shared returns); **no "interest" in a scheme** (a bilateral credit contract with a single lender, whose capital sits in an offset account, not a fund or trust); and **no expectation of financial return to the borrower** (the borrower is a consumer receiving credit for retirement income, not an investor receiving distributions). The EPM is economically equivalent to a reverse mortgage with a different repayment mechanism, not an investment.

The EPM meets the statutory definition of a credit contract

A credit contract exists when credit is provided, a charge is made for it, the debtor is a natural person, and the credit is predominantly for personal/domestic purposes.

The EPM satisfies all four:

- The lender advances credit (and holds back a portion of principal via the offset account)
- The lender earns margin; the borrower is a natural person
- The purpose is retirement income or aged-care funding.

The index-linked repayment mechanism does not change its legal character — credit contracts can have variable, index-linked or contingent repayment structures, without becoming an MIS.

The security structure reinforces credit, not investment

The EPM is secured by a registered first mortgage, a linked offset account holding the lender's capital, and a run-off mechanism that repays the lender from the mortgage itself.

This is textbook secured lending: no trust, no pooling, no unitization, no investor contributions, no investment manager, no fiduciary obligations, no collective enterprise. The lender is a secured creditor, not a scheme operator.

Regulator guidance and economic substance

ASIC has repeatedly stated that credit products are not MISs even when they involve contingent repayment, property security, indexation or complex mechanics. The key distinction is who takes the financial risk and why: in the EPM the lender takes credit risk, the borrower receives credit, and the borrower is not seeking investment returns — the opposite of an MIS. On economic substance, the EPM provides predictable income, uses home equity as security, avoids compounding interest, allows early termination without penalty, lets the borrower retain full ownership and title, and repays itself through the mortgage offset mechanism. Nothing here resembles an investment scheme; everything resembles a regulated credit contract.

Conclusion: The EPM is a secured credit contract — it provides credit to a consumer, secured by a registered mortgage, with repayment sourced from the mortgage structure itself. The borrower does not deposit monies or contribute assets, does not acquire an interest in a pooled arrangement, and does not expect a financial return. Accordingly, the EPM does not satisfy the statutory definition of a managed investment scheme (s9 Corporations Act).

15. Methodology

Key Assumptions

Component	Model	Parameters
Equity returns	GBM with Stochastic Drift + Mean Reversion (quarterly)	mu=9.2%, sigma=16.6%, gamma=0.163 (after asymmetric collar)
Asymmetric hedge collar	Cap/Floor on returns	Cap: +40%, Floor: -20% per period
Cash rate	OU mean-reversion	theta=2.13%, kappa=0.24, sigma=1.22%
Equity-rate correlation	0.30	Appropriate for 30yr horizon (long-term marginally positive)
House prices	Deterministic (by design)	Sets LVR at origination only; not a shared appreciation mortgage
Prepayment/Early exit	Run-off mechanism	No exit until all loan costs paid from mortgage offset account
Hedging cost	Calculated put-call	+0.046% p.a. (cap 1.4 / floor 0.8)
Variable costs	Fixed spread	3.45% (includes LMI + tail risk premiums)
P&I; amortisation	Linear PI	Builds to $1.2M peak, then amortising repayment to Yr 30
Insurance discount	2.13%	Cash rate mean; claims only at mortgage expiry
LMI coverage	80% of deficit paths	Worst 20% of deficit paths (larger losses) = tail risk (reinsurance)
Payments & Risk Waterfall	Portfolio level	ETF selldown → cross-subsidise → net deficit → PoC
Surplus split	50/50	Futureproof and Mortgage Funder at maturity
Profit realisation	Every 3 years	10% of running surplus drawn at each reset
Simulation paths	50,000	SE 0.12% on PoD

Metrics Definitions

PoD (Probability of Deficit): Fraction of paths where investment balance < mortgage balance at a given point in time. A balance sheet snapshot — useful but NOT the probability of an insurance claim.

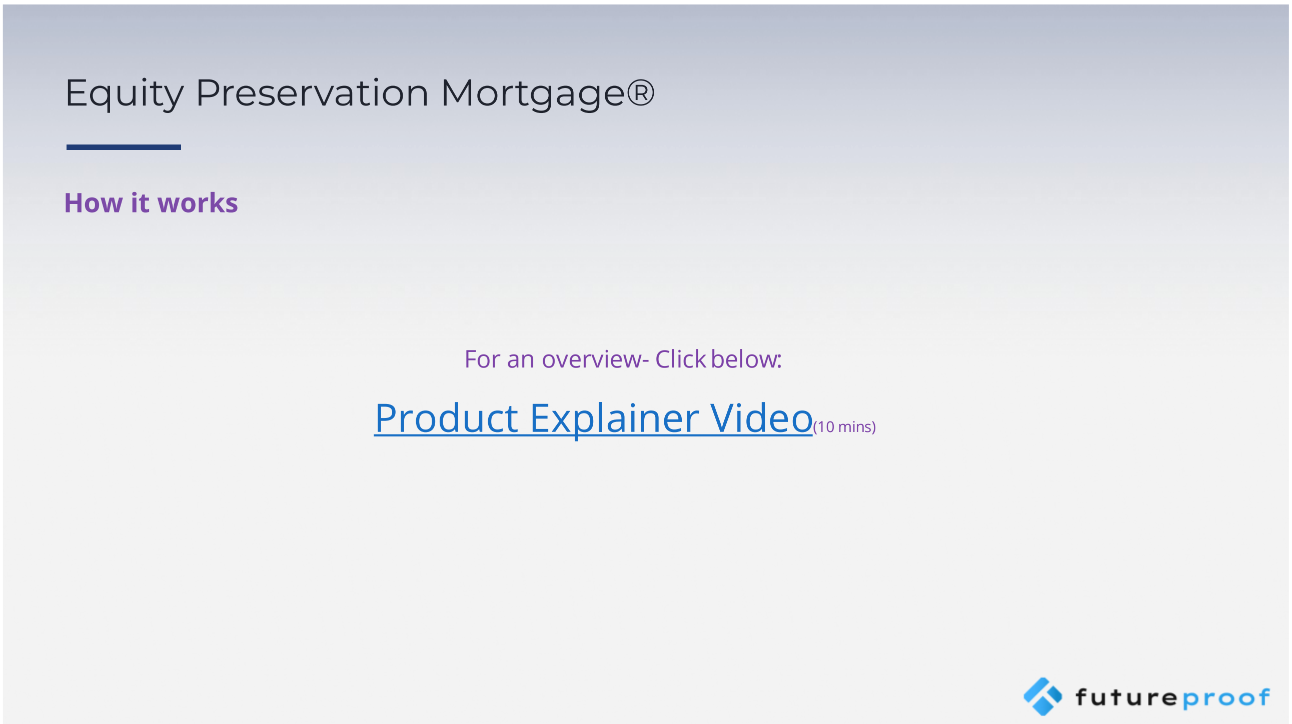
PoC (Probability of Claim): Probability of an actual insurance claim on an expiring mortgage, after the Payments & Risk Waterfall has been applied. This is the key metric for insurance and reinsurance pricing. Claims can only occur at mortgage expiry.

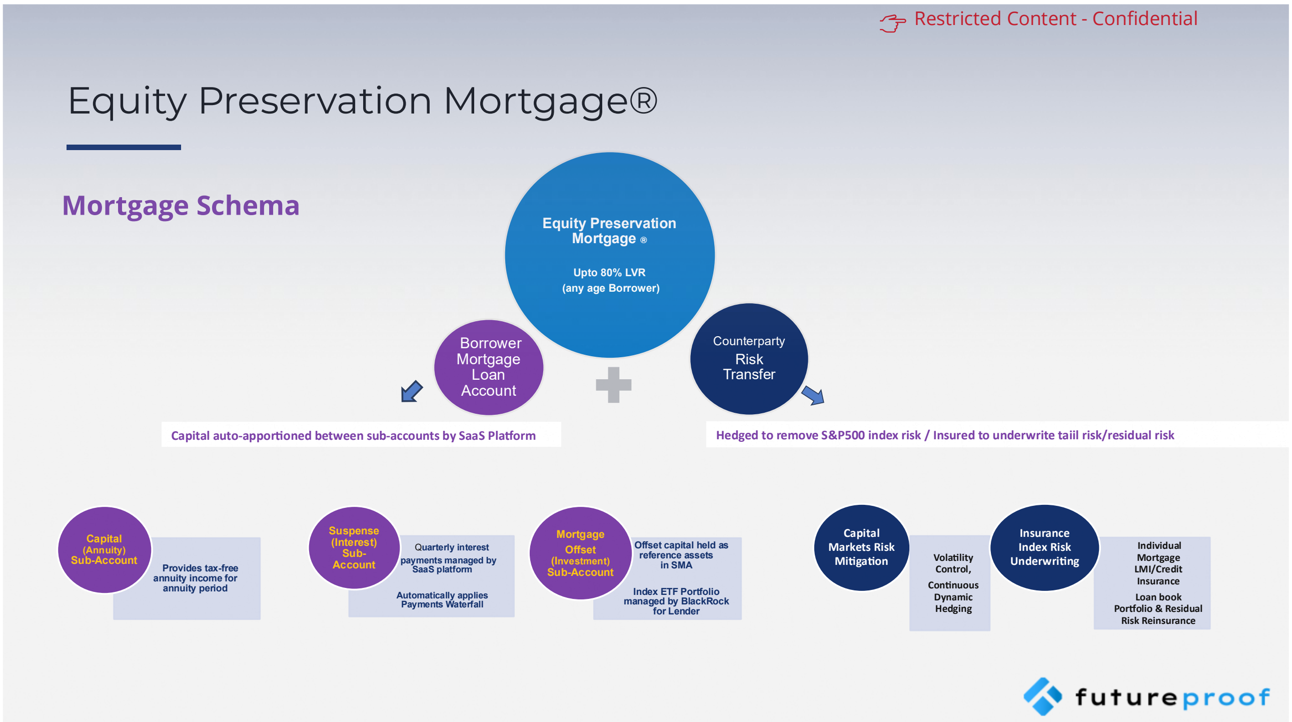
Payments & Risk Waterfall: Portfolio-level mechanism: (1) sell down ETFs, (2) cross-subsidizes from surplus loans, (3) only then determine net deficit and PoC.

Discounted Expected Deficit (S3): The present value of expected loss, discounted at 2.13% (stochastic cash rate mean). Basis for insurance premium calculation.

Appendix A – EPM Design

Mortgage Schema





Appendix B – EPM Credit Risk Management

Payments & Risk Waterfall